# COMPACT OBJECTS FORM A KAROUBIAN SATURATED TRIANGULATED SUBCATEGORY

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 11371-11383, following the statement at 11363-11369.

*Editorial note.* The five-lemma step compares the long exact sequence of the given triangle with the direct sum of its coordinate sequences; both tools are named and no new triangle is constructed.

*Allowed standard prerequisites (editorial).*

- Representable functor is homological. lemma-representable-homological (carded L26235, cited at ~11373). The long exact Hom sequences compared by the five lemma.
- Five lemma and Karoubi images. Homology, Lemma homology-lemma-five-lemma (cited at ~11375) and lemma-projectors-have-images-triangulated (same chapter, cited at ~11381). Cone closure of compactness; Karoubianness transfers to the subcategory.

*Source reference.* Original reference 05QB: focused-seeds/geometry/S9-der-compact-objects-subcategory/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 05QW: sources/stacks-derived/pinned-original-source.tex, source lines 1023–1023. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{D}$ be a (pre-)triangulated category with direct sums. Then the compact objects of $\mathcal{D}$ form the objects of a Karoubian, saturated, strictly full, (pre-)triangulated subcategory $\mathcal{D}_c$ of $\mathcal{D}$.*

*Proof.* Let $(X, Y, Z, f, g, h)$ be a distinguished triangle of $\mathcal{D}$ with $X$ and $Y$ compact. Then it follows from Lemma 0149 and the five lemma (Homology, Lemma 05QB) that $Z$ is a compact object too. It is clear that if $X \oplus Y$ is compact, then $X$, $Y$ are

*Date*: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

compact objects too. Hence $\mathcal{D}_c$ is a saturated triangulated subcategory. Since $\mathcal{D}$ is Karoubian by Lemma 05QW we conclude that the same is true for $\mathcal{D}_c$. $\square$